

Technical Report

AI-Powered Image Classification Platform

End-to-end deep-learning image classification using transfer learning, REST API, Streamlit, and prediction history.

1. Objective

Develop a production-oriented image classification platform that trains and compares multiple deep-learning models, evaluates them using standard classification metrics, exposes image prediction through a REST API, and provides a simple browser-based interface.

2. Dataset

CIFAR-10 is used as the public dataset. It contains 60,000 32×32 RGB images across ten classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck.

Official source: <https://www.cs.toronto.edu/~kriz/cifar.html>

3. Data Pipeline

```
CIFAR-10
|
+--> Train / Validation / Test split
|
+--> Resize to 224x224
|
+--> Training augmentation
|   - Random horizontal flip
|   - Random crop
|   - Color jitter
|
+--> ImageNet normalization
|
+--> Transfer learning
|   |-- ResNet18
|   `-- MobileNetV3-Small
```

The demonstrated training run used 10,000 training images and 2,000 validation images. Final evaluation was performed on the full 10,000-image CIFAR-10 test set.

4. Model Architecture and Training Strategy

ResNet18: ImageNet-pretrained residual network with a new ten-class classification layer.

MobileNetV3-Small: ImageNet-pretrained lightweight convolutional network with a new ten-class classifier.

For the demonstrated run, the convolutional backbone was frozen and the classification head was fine-tuned using AdamW with learning rate 1e-3, weight decay 1e-4, and 5 epochs. Training ran on an NVIDIA GeForce GTX 1650 using CUDA.

5. Performance Comparison

Model	Best validation accuracy	Test accuracy (2k demo subset)
ResNet18	78.10%	—
MobileNetV3-Small	81.45%	80.65%

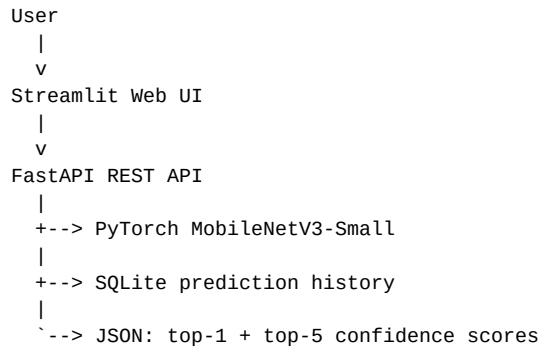
MobileNetV3-Small was selected because it achieved the strongest validation accuracy in the demonstrated run. Its final model was then evaluated on the full CIFAR-10 test set.

6. Final Full-Test Evaluation

Metric	Result
Accuracy	80.93%
Precision (weighted)	81.74%
Recall (weighted)	80.93%
F1-score (weighted)	80.86%
Top-5 accuracy	99.28%

The confusion matrix is generated by the evaluation pipeline and stored as reports/confusion_matrix.png.

7. Deployment Architecture



The REST service exposes /health, /classes, /history, and /predict. The Streamlit frontend provides image upload, classification, confidence scores, and prediction history.

8. Demonstrated Prediction

A CIFAR-10 test image with ground-truth class **cat** was uploaded through the application. The model correctly predicted **cat** with **85.5%** confidence. The top-five output also displayed dog, bird, frog, and airplane with lower confidence.

9. Validation and Testing

Automated API tests were executed with pytest: **2 passed**. Four console warnings were deprecation warnings from the FastAPI/Starlette testing stack and did not cause failures.

10. Deliverables

Source code, training scripts, experiment notes, trained PyTorch model, evaluation artifacts, REST API, Streamlit frontend, Docker configuration, README documentation, and technical report are included in the project package.

11. Conclusion

The platform demonstrates the complete workflow from public data ingestion and augmentation through transfer learning, model comparison, quantitative evaluation, REST deployment, browser-based inference, and persistent prediction history. MobileNetV3-Small achieved 80.93% accuracy and 99.28% top-5 accuracy on the full CIFAR-10 test set for the demonstrated training configuration.